


Lecture 5: Unsupervised Learning
PCA, t-SNE, Clustering, Autoencoders
Blood Cells, Credit-Card Fraud and Tumour Genes without Label

# Deep Learning for Scientific Discovery

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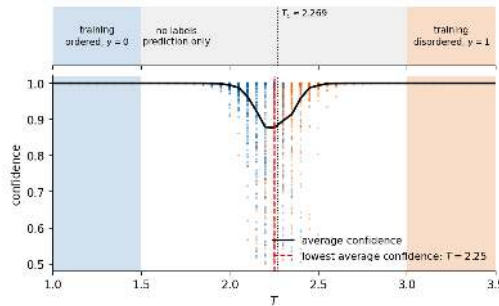
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Natural Sciences
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# Recap: Finding $T_c$ with Labels

- a CNN trained on spin configurations labelled "ordered" ($T < 1.5$) and "disordered" ($T > 3$)
- its average confidence is lowest at $T = 2.25$ ($L = 40$); exact value $T_c \approx 2.269$
- but the labels came from us: we knew that there are two phases, and where each of them lies




**Today: what can we learn from data without labels?**

# 1

# Learning without Labels



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# Unsupervised Learning

Supervised learning needs labels, and labels are expensive (often an expert labels every sample), they contain what we already know (the ten digits, the two phases of a magnet), and in discovery the categories are often unknown.

## Supervised

data: pairs $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$

goal: a function $f(\mathbf{x}) \approx y$, i.e. learn $p(y \mid \mathbf{x})$

the question is fixed in advance: which cell type? which phase?

## Unsupervised

data: samples only, $\{\mathbf{x}_i\}_{i=1}^n$

goal: learn about $p(\mathbf{x})$, the distribution the data come from

no question fixed in advance: we want to **understand the data**

# Understanding the Data

| question about the data | method |
|---|---|
| which few variables describe a sample? | dimensionality reduction |
| how many kinds of samples are there? | clustering |
| what is typical, and what is unusual? | anomaly detection |
| how are the samples distributed? | density estimation, |
| can we draw new ones? | generative models |

**The method proposes a structure; deciding what it means is ours.**

# High-Dimensional Data

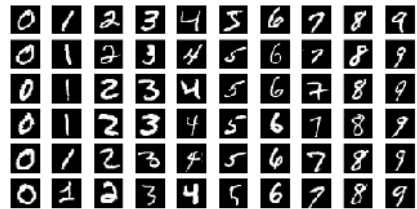
every sample is a vector $\mathbf{x} \in \mathbb{R}^D$: an MNIST digit has $D = 28 \cdot 28 = 784$ pixels, an Ising configuration $D = L^2$ spins

We cannot look at 784 dimensions. But the data do not fill $\mathbb{R}^{784}$: pick 784 random pixel values and you get noise, never a digit. Real images lie in a thin, low-dimensional part of the space.

So the analysis starts with **dimensionality reduction**: project the data on a smaller space,

$$\mathbf{x} \in \mathbb{R}^D \;\mapsto\; \mathbf{z} \in \mathbb{R}^d, \qquad d \ll D,$$

keeping their structure; then look at the points $\mathbf{z}_i$: do they form groups?



# 2

# Principal Component Analysis



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# Principal Component Analysis: the Covariance Matrix

Data $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^D$; $x_{ij}$ is feature $j$ (e.g. pixel $j$) of sample $i$. The mean $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i$ (for MNIST: the average image), and the **covariance matrix** of the features, a $D \times D$ matrix:

$$C_{jl} = \frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l), \qquad j, l = 1, \ldots, D$$

- $C_{jj}$: the variance of feature $j$
- $C_{jl}$: how features $j$ and $l$ vary together over the data ($> 0$: rise together; $< 0$: one rises, the other falls); with each feature divided by its standard deviation, $C$ becomes the *correlation* matrix

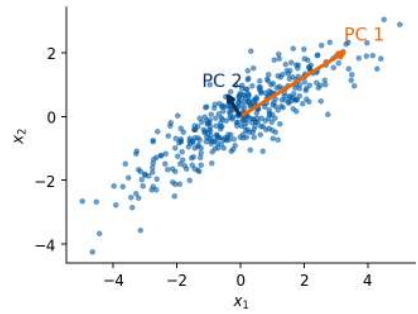
K. Pearson, *On lines and planes of closest fit to systems of points in space*, Phil. Mag. **2**, 559 (1901), doi:10.1080/14786440109462720; H. Hotelling, *Analysis of a complex of statistical variables into principal components*, J. Educ. Psychol. **24**, 417 (1933), doi:10.1037/h0071325

# Principal Component Analysis: a Linear Projection

- diagonalise $C$: $C\,\mathbf{v}_a = \lambda_a \mathbf{v}_a$, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_D \geq 0$; the variance of the data along a unit vector $\mathbf{u}$ is $\mathbf{u}^\top C\,\mathbf{u}$, largest for $\mathbf{u} = \mathbf{v}_1$ (value $\lambda_1$)
- keep the $m$ eigenvectors with the largest eigenvalues, the **principal components**
- project every sample on them:

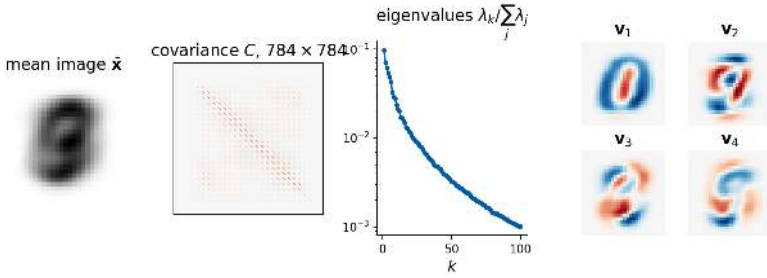
$$z_{ia} = \mathbf{v}_a \cdot (\mathbf{x}_i - \bar{\mathbf{x}}), \qquad a = 1, \ldots, m$$

- in matrix form $\mathbf{z} = V^\top(\mathbf{x} - \bar{\mathbf{x}})$, $V = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ the $D \times m$ matrix of the kept eigenvectors: a linear map



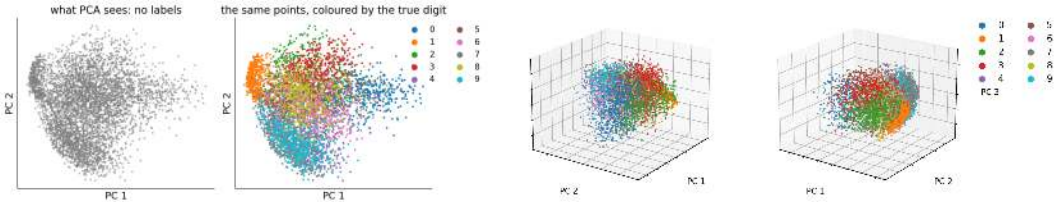

400 correlated points: PC 1 carries 94.5% of the variance

# PCA on MNIST: Covariance and Eigenvectors




- 60 000 training images of 784 pixels; $C$ is $784 \times 784$: how two pixels vary together over the images
- the eigenvalues fall fast: 2 components keep 17% of the variance, 10 keep 49%, 87 keep 90%
- the eigenvectors are themselves images: patterns of pixels that vary together

# PCA on MNIST




- 5000 test digits projected on the first two components (left): they keep 17% of the variance; three components (right, two viewing angles) keep 23%; 50 components keep 82%
- is there clustering? some digits form separate groups (0, 1), the third direction separates a few more, but most overlap: the digits do not lie in a flat, low-dimensional subspace, and a linear projection is too crude

# 3
# t-SNE



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# t-SNE: Similarities in the Data

**Neighbours in $\mathbb{R}^D$ stay neighbours in the 2D map**; the map positions are the parameters, found by gradient descent (no network). Instances $\mathbf{x}_i \in \mathbb{R}^D$, $i = 1, \ldots, n$ (an MNIST image: $D = 784$), squared distance $\|\mathbf{x}_i - \mathbf{x}_j\|^2 = \sum_{k=1}^{D} (x_{ik} - x_{jk})^2$. For instance $i$, a Gaussian of width $\sigma_i$ turns distances into neighbour probabilities over the $n - 1$ other instances:

$$p_{j|i} = \frac{e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2}}{\sum_{l \neq i} e^{-\|\mathbf{x}_i - \mathbf{x}_l\|^2 / 2\sigma_i^2}}, \qquad p_{i|i} = 0, \qquad p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$$

- $p_{j|i}$: the probability that $\mathbf{x}_i$ picks $\mathbf{x}_j$ as its neighbour; $\sigma_i$ is chosen so that the **perplexity** $2^{H_i}$, $H_i = -\sum_j p_{j|i} \log_2 p_{j|i}$, equals a set value (here 30): the effective number of neighbours ($k$ equally likely neighbours give $2^{H_i} = k$)
- $P = \{p_{ij}\}$, $\sum_{i \neq j} p_{ij} = 1$: a probability distribution over pairs, how likely $i$ and $j$ are neighbours *in the original space*

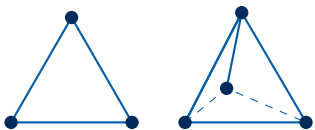
# t-SNE: Similarities in the Map

$\mathbf{z}_i \in \mathbb{R}^d$: the point that represents instance $i$ in a map of dimension $d$ we choose ($d = 2$ to plot it); the $\mathbf{z}_i$ are the unknowns.

$$q_{ij} = \frac{(1 + \|\mathbf{z}_i - \mathbf{z}_j\|^2)^{-1}}{\sum_{l \neq m}(1 + \|\mathbf{z}_l - \mathbf{z}_m\|^2)^{-1}}, \qquad \|\mathbf{z}_i - \mathbf{z}_j\|^2 = \sum_{a=1}^{d}(z_{ia} - z_{ja})^2$$

- the numerator is a Student-t kernel of the map distance; in the original space the kernel was a Gaussian
- the denominator sums over all pairs, so $\sum_{i \neq j} q_{ij} = 1$: $Q = \{q_{ij}\}$ is the same kind of distribution over pairs, *in the map*
- t-SNE moves the $\mathbf{z}_i$ until $Q$ matches $P$: pairs that are neighbours in $\mathbb{R}^D$ (large $p_{ij}$) must get large $q_{ij}$, i.e. be close in the map

# The Crowding Problem



plane: at most 3 3D: 4

points all at the *same* distance from each other: $D+1$ in $\mathbb{R}^D$, so 785 in the MNIST pixel space, but only 3 in a 2D map

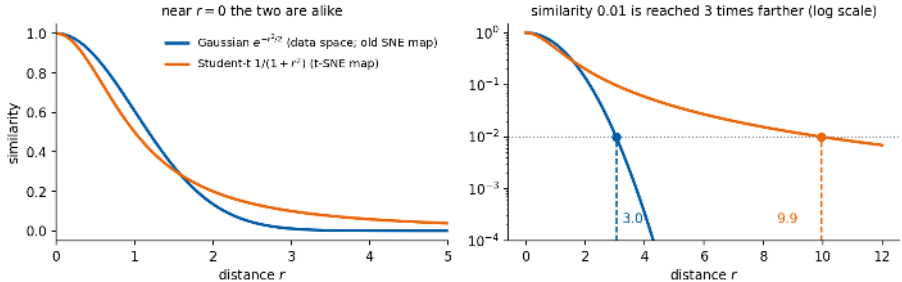
**Too little room.** Groups roughly equally far apart in $\mathbb{R}^{784}$ (the ten digits) cannot all be equally far apart in the plane: some pairs at a moderate distance must be placed much farther away.

**With a Gaussian in the map** (the older SNE): a pair placed far away gets $q_{ij} \approx 0$, although its $p_{ij}$ is small but not zero. The cost pulls such pairs back, and all groups are squeezed into the centre of the map: the *crowding problem*.

L. van der Maaten, G. Hinton, *Visualizing Data using t-SNE*, J. Mach. Learn. Res. **9**, 2579 (2008),

jmlr.org/papers/v9/vandermaaten08a

# The Heavy Tail of the Student-t




The Student-t decays like $1/r^2$, the Gaussian like $e^{-r^2/2}$. A small similarity such as 0.01 is reached at $r = 3.0$ with the Gaussian but at $r = 9.9$ with the Student-t. So a moderately distant pair (small $p_{ij}$) can be placed far apart in the map and still get a matching $q_{ij}$: there is room to push groups apart, and **gaps open between them**.

L. van der Maaten, G. Hinton, *Visualizing Data using t-SNE*, J. Mach. Learn. Res. **9**, 2579 (2008), jmlr.org/papers/v9/vandermaaten08a

# t-SNE: the Loss

Two distributions over pairs of instances: $P$ from the original space, $Q$ from the map. Move the map points $\mathbf{z}_i$ by gradient descent until $Q$ matches $P$, i.e. minimise their Kullback–Leibler divergence:

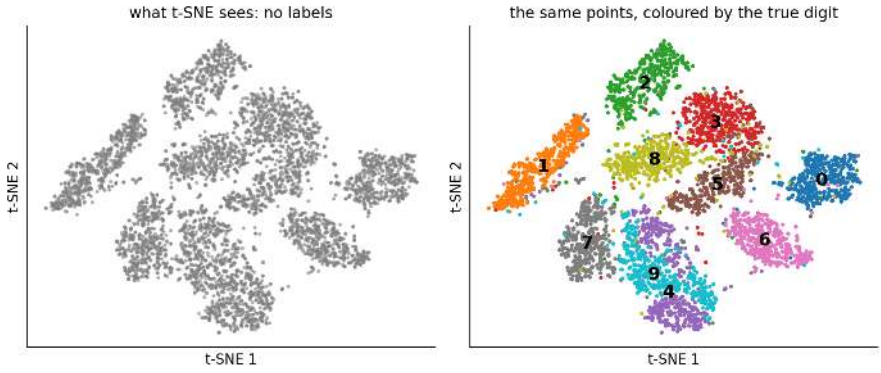
$$\mathcal{L}(\mathbf{z}_1, \ldots, \mathbf{z}_n) = \mathrm{KL}(P \,\|\, Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

- neighbours placed far apart ($p_{ij}$ large, $q_{ij}$ small): large cost
- distant instances placed close ($p_{ij}$ small): small cost
- so t-SNE keeps the **local** structure; distances between groups in the map carry little meaning

**The result is a set of 2D points, one per instance, that we can plot and inspect.**

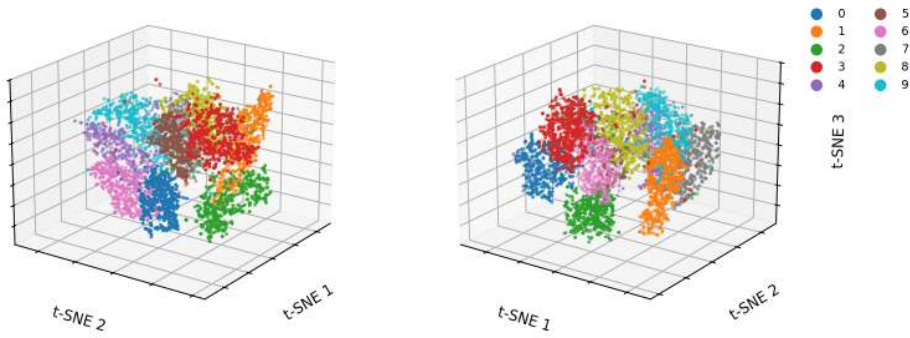
L. van der Maaten, G. Hinton, *Visualizing Data using t-SNE*, J. Mach. Learn. Res. **9**, 2579 (2008), jmlr.org/papers/v9/vandermaaten08a

# t-SNE on MNIST



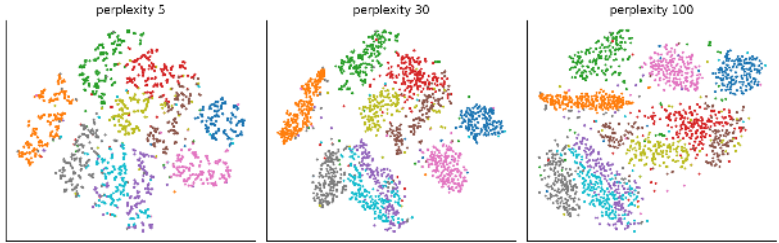

- 5000 test digits, $d = 2$; the labels only colour the right panel
- ten groups appear without any label

# t-SNE on MNIST in 3D




The same loss with $\mathbf{z}_i \in \mathbb{R}^3$: the 5000 digits seen from two angles.

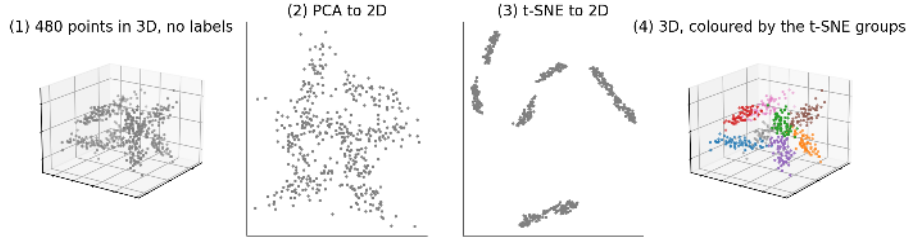
# The t-SNE Map




2000 digits at three perplexities: small perplexity keeps only very close neighbours, large perplexity smooths the map

- **meaningless:** the position of a cluster (a rotated, mirrored or shifted map has the same loss); the separation between clusters (distant pairs have $p_{ij} \approx 0$); cluster sizes ($\sigma_i$ adapts to the density)
- **reliable:** which instances are close neighbours

# t-SNE on Data in 3D




- 480 points in $\mathbb{R}^3$, *without* labels (1): eight elongated groups passing through each other; neither the 3D view nor PCA (2) shows how many there are
- t-SNE keeps neighbours together: its 2D map (3) shows eight separate groups; coloured back in 3D (4), they are the groups that generated the data

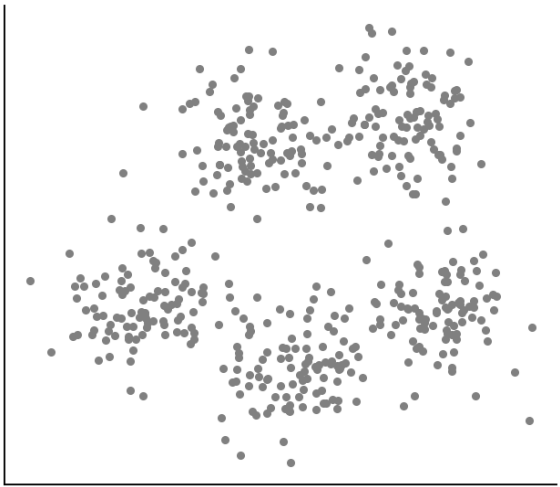
# 4

# Clustering



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# Points without Labels



500 points in 2D, no labels.

*We see some groups. How many are there, and where is the boundary between two of them?*

# k-means Clustering

**Goal:** split the $n$ instances $\mathbf{x}_1, \dots, \mathbf{x}_n$ into $k$ clusters $S_1, \dots, S_k$ (sets of instance indices) with centres $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_k$, minimising the total squared distance of the instances to their centres:

$$\mathcal{L} = \sum_{c=1}^{k} \sum_{i \in S_c} \|\mathbf{x}_i - \boldsymbol{\mu}_c\|^2$$

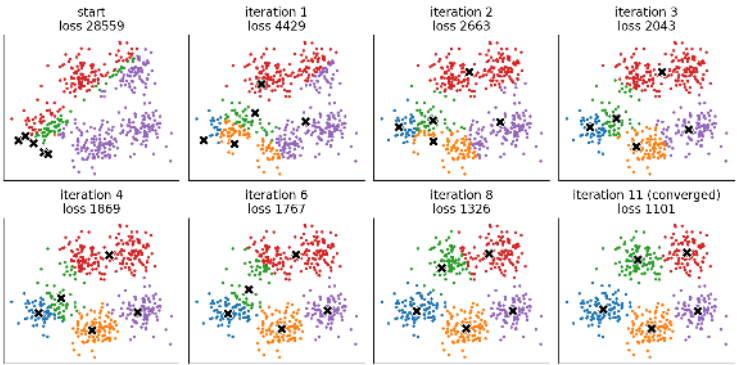
**Algorithm:** start from $k$ centres; repeat until no instance changes cluster:

- *assign* each instance to its nearest centre:
  $S_c = \{\, i : \|\mathbf{x}_i - \boldsymbol{\mu}_c\| \le \|\mathbf{x}_i - \boldsymbol{\mu}_{c'}\| \text{ for all } c' \,\}$
- *update* each centre to the mean of its instances: $\boldsymbol{\mu}_c = \frac{1}{|S_c|} \sum_{i \in S_c} \mathbf{x}_i$

No move raises $\mathcal{L}$; keep the best of several random starts. $k$ is chosen by us.

S. P. Lloyd, *Least squares quantization in PCM*, IEEE Trans. Inf. Theory **28**, 129 (1982), doi:10.1109/TIT.1982.1056489

# k-means Iterations

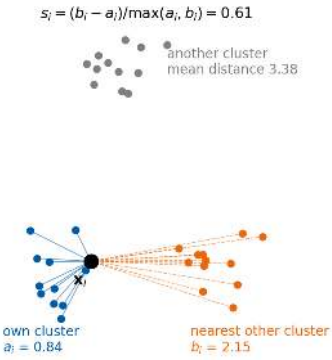



the same 500 points, $k = 5$

start: five centres (crosses) placed badly, at the five points nearest the lower-left corner; colours: the assignment

the loss falls from 28 559 to 1 101; after 11 iterations nothing changes, and the clusters match the five groups that generated the points (ARI 0.93)

# The Silhouette Score




In a good clustering each instance is close to its own cluster and far from the others. For $\mathbf{x}_i$ in cluster $S_c$:

- $a_i$: mean distance to the other instances of $S_c$
- $b_i$: mean distance to the instances of the nearest other cluster

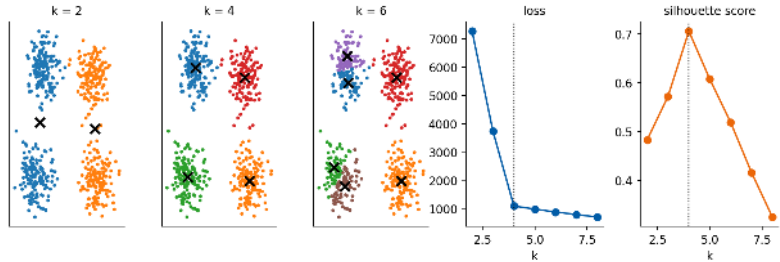
$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)} \in [-1, 1]$$

$s_i \approx 1$: well inside its cluster; $s_i \approx 0$: on a border; $s_i < 0$: probably in the wrong cluster.

Mean silhouette $\bar{s}(k) = \frac{1}{n}\sum_i s_i$ for the clustering with $k$ clusters: choose the $k$ with the largest $\bar{s}(k)$.

P. J. Rousseeuw, *Silhouettes: A graphical aid to the interpretation and validation of cluster analysis*, J. Comput. Appl. Math. **20**, 53 (1987), doi:10.1016/0377-0427(87)90125-7

# Choosing the Number of Clusters




Four groups, so the right answer is known. For $k = 2, \ldots, 8$: run k-means (10 random starts), record the loss $\mathcal{L}(k)$ and the mean silhouette $\bar{s}(k)$.

- **elbow:** $\mathcal{L}(k)$ always decreases with $k$; fast up to the right $k$, slowly after it: the bend is at $k = 4$
- **silhouette:** $\bar{s}(k)$ is highest at $k = 4$; too small $k$ merges groups, too large $k$ splits them

# Rand Index

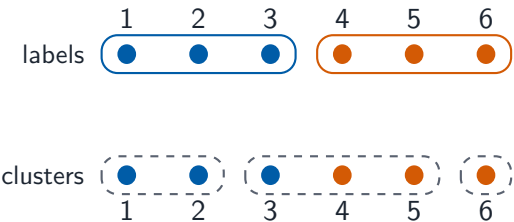
Compare a clustering with known classes *pair by pair*; the names of the groups do not matter.

A pair of instances $(i, j)$ **agrees** if both partitions put it together, or both put it apart. Out of all $n(n-1)/2$ pairs:

- $a$: pairs together in both
- $b$: pairs apart in both

$$\mathrm{RI} = \frac{a+b}{n(n-1)/2} \in [0,1]$$

$\mathrm{RI} = 1$: identical partitions; every disagreeing pair lowers it.



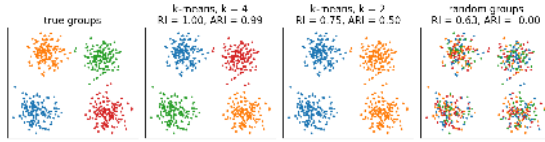

together in both: 1–2, 4–5: $a = 2$

apart in both: the other 7 pairs: $b = 7$

together in one only (disagree):
1–3, 2–3, 3–4, 3–5, 4–6, 5–6

$$\mathrm{RI} = (2+7)/15 = 0.60$$

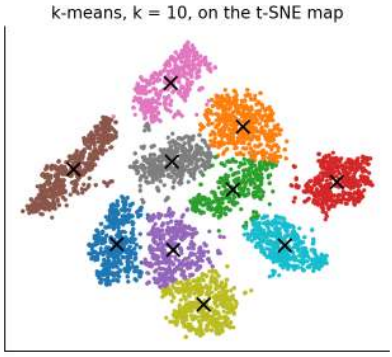
# Adjusted Rand Index



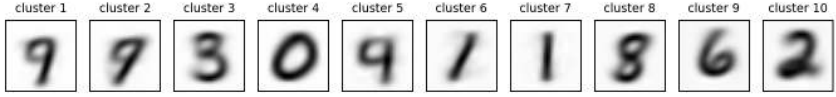

- random groups already reach $\mathrm{RI} = 0.63$: most pairs are apart in both partitions
- the adjusted Rand index subtracts what chance gives, $\mathrm{ARI} = \dfrac{\mathrm{RI} - \mathbb{E}[\mathrm{RI}]}{1 - \mathbb{E}[\mathrm{RI}]}$; $\mathbb{E}[\mathrm{RI}]$: the mean $\mathrm{RI}$ when the instances are reassigned at random, keeping the group sizes
- **ARI = 1**: the same partition; **ARI $\approx$ 0**: no better than chance (it can be negative); the six points of the sketch: 0.12
- the eight groups in 3D, k-means ($k = 8$): ARI 0.46 on the PCA projection, 0.96 on the t-SNE map

L. Hubert and P. Arabie, *Comparing partitions*, J. Classification **2**, 193 (1985), doi:10.1007/BF01908075

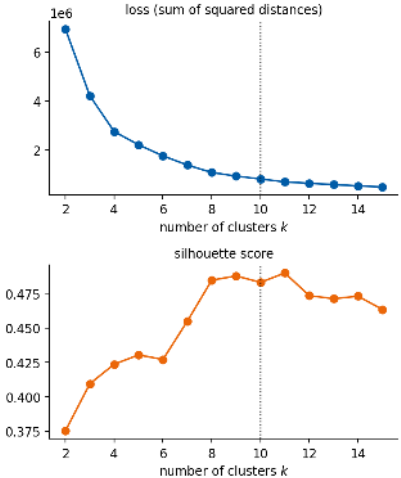
# k-means on MNIST




- $k = 10$, the number of digits; 5000 test digits
- agreement with the true digits, ARI: **0.38** on 50 PCA components, **0.76** on the t-SNE map
- below: the ten centres $\boldsymbol{\mu}_c$ found on 50 PCA components, shown as images

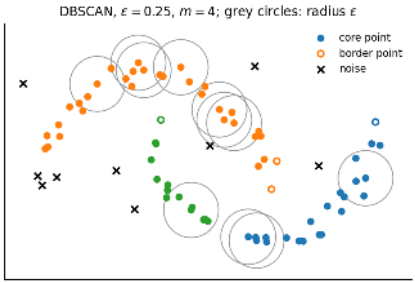


# Choosing k on MNIST




- the loss $\mathcal{L}(k)$ and the mean silhouette $\bar{s}(k)$ on the t-SNE map of the 5000 digits
- dotted line: $k = 10$, the number of digits, drawn only for comparison
- no sharp elbow; the silhouette is highest at $k = 11$ (one digit split in two)
- on real data the guides suggest a range of $k$; the final choice uses what we know about the data

# DBSCAN




*Density-Based Spatial Clustering of Applications with Noise*. Parameters: a radius $\varepsilon$, a minimum number of points $m$; the $\varepsilon$-neighbourhood of instance $i$ (itself included):

$$N_\varepsilon(i) = \{\, j : \|\mathbf{x}_j - \mathbf{x}_i\| \leq \varepsilon \,\}$$

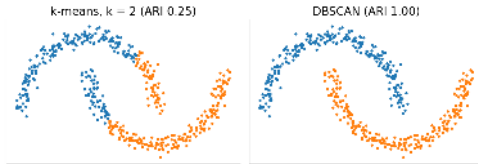
- **core point:** $|N_\varepsilon(i)| \geq m$ (grey circles)
- **border point:** not core, but $i \in N_\varepsilon(j)$ for some core point $j$
- **noise:** all other points (crosses)

A cluster: core points linked by chains of core neighbours, plus their border points.

M. Ester, H.-P. Kriegel, J. Sander, X. Xu, *A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise*, KDD 1996, AAAI proceedings

# DBSCAN: the Algorithm

1. compute $N_\varepsilon(i)$ for every instance and mark the core points
2. take a core point not yet in a cluster and start a new cluster with it
3. add every point of $N_\varepsilon(j)$ for every core point $j$ of the cluster; repeat until no point is added: the cluster grows through the dense region and stops at its border points
4. repeat 2 and 3 until every core point is in a cluster; points never reached are noise




k-means cuts the space into convex cells, so it finds round clusters: on two half-moons k-means ($k = 2$) gives ARI 0.25, DBSCAN 1.00. DBSCAN needs no number of clusters, and clusters can have any shape.

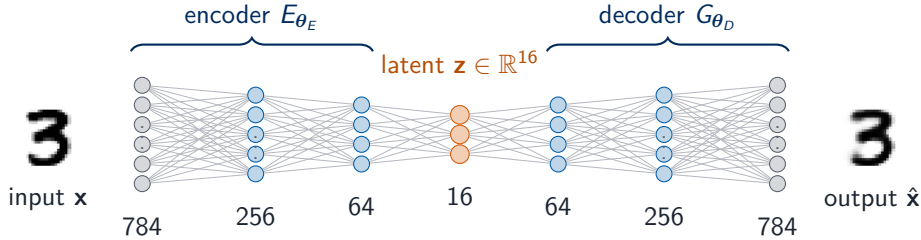
# 5
# Autoencoders



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# Autoencoder

If a network can rebuild an image from only $d$ numbers, those numbers contain what is essential about it (PCA needs 87 components for 90% of the MNIST variance).




- **encoder**, a network $\mathbb{R}^D \to \mathbb{R}^d$ with parameters $\boldsymbol{\theta}_E$: $\mathbf{z} = E_{\boldsymbol{\theta}_E}(\mathbf{x})$; **decoder**, $\mathbb{R}^d \to \mathbb{R}^D$ with $\boldsymbol{\theta}_D$: $\hat{\mathbf{x}} = G_{\boldsymbol{\theta}_D}(\mathbf{z})$
- the bottleneck $d \ll D$ forces $\mathbf{z}$ to keep the features the decoder needs: a learned **latent-space representation**

G. E. Hinton, R. R. Salakhutdinov, Science **313**, 504 (2006), doi:10.1126/science.1127647

# The Autoencoder Loss




Each input $\mathbf{x}_i$ is compared with its reconstruction $\hat{\mathbf{x}}_i = G_{\theta_D}(E_{\theta_E}(\mathbf{x}_i))$, pixel by pixel:

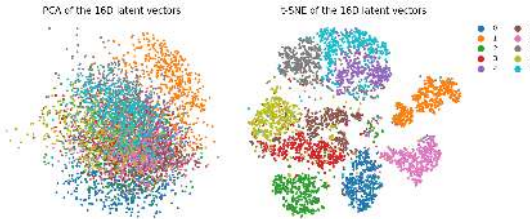
$$\mathcal{L}(\boldsymbol{\theta}_E, \boldsymbol{\theta}_D) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{D}\sum_{j=1}^{D}\left(x_{ij} - \hat{x}_{ij}\right)^2$$

$i$: image, $j$: pixel (right panel: one digit). No labels: the target is the input itself.

- trained by gradient descent; MNIST: dense layers $784 \to 256 \to 64 \to d$ and back
- a linear autoencoder with squared loss finds the PCA subspace (Baldi & Hornik 1989); non-linear layers can follow curved structure

P. Baldi, K. Hornik, *Neural networks and principal component analysis: Learning from examples without local minima*, Neural Networks **2**, 53 (1989), doi:10.1016/0893-6080(89)90014-2

# Compression and the Latent Space




5000 test digits, $d = 16$ latent numbers each, drawn in 2D; colours: true digits, used only for plotting

- **compression:** 784 pixels become 16 numbers $\mathbf{z} = E_{\theta_E}(\mathbf{x})$; the encoder is a continuous function, so similar images get nearby $\mathbf{z}$, and the bottleneck keeps only what the decoder needs
- **PCA of z** overlaps the digits; **t-SNE of z** keeps the neighbours in 16D and shows the groups

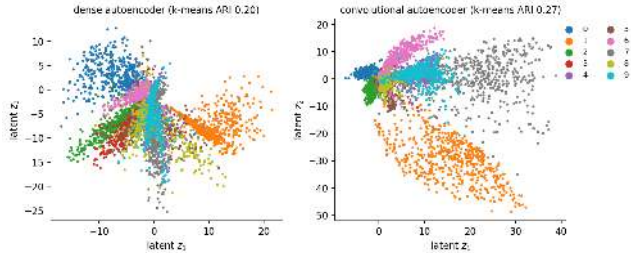
# k-means in the Latent Space

k-means ($k = 10$) against the true digits, adjusted Rand index:

| | directly | on the t-SNE map |
|---|---|---|
| pixels (50 PCA components) | 0.38 | 0.76 |
| 16 latent numbers of the autoencoder | 0.28 | 0.67 |

- the digits are kept in the latent space, but not as the round groups that k-means looks for
- t-SNE only *draws* the latent space; the latent space itself takes *new* images (encoder) and gives back images (decoder)

# Dense and Convolutional Autoencoders




- $d = 2$ latent numbers, so the latent space is drawn directly; colours: true digits
- **dense** (left): a flat list of 784 pixels; **convolutional** (right): two convolutions with stride 2 ($28 \to 14 \to 7$ pixels, 32 and 64 channels)
- k-means ($k = 10$), ARI: 0.20 (dense), 0.27 (convolutional); in this run (one seed) the convolutional encoder, which compares neighbouring pixels, separates the digits better

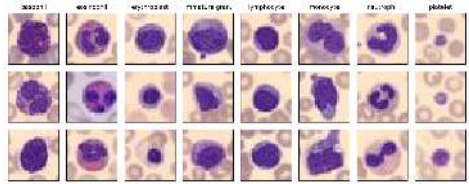
# 6

# Blood Cells without Labels



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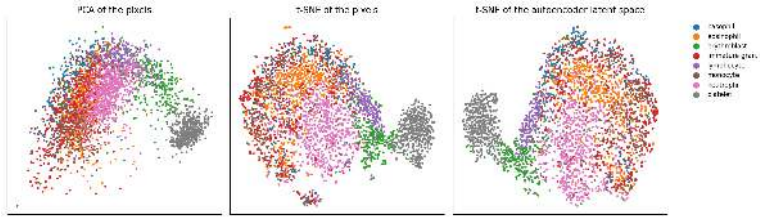
# BloodMNIST



- microscope images of single *normal* blood cells, eight types assigned by clinical pathologists; $28 \times 28$ pixels, 3 colours: $D = 2352$; 11 959 training and 3421 test images
- white blood cells: basophil, eosinophil, neutrophil, lymphocyte, monocyte, immature granulocyte (not yet mature); erythroblast: an immature red blood cell; platelet: a small cell fragment

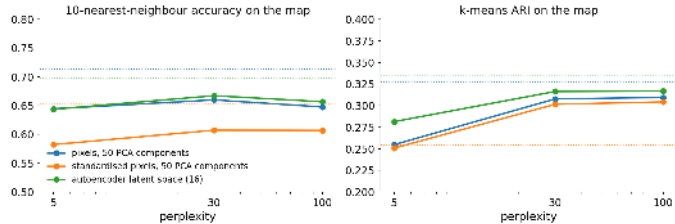
A. Acevedo et al., *A dataset of microscopic peripheral blood cell images for development of automatic recognition systems*, Data in Brief **30**, 105474 (2020), doi:10.1016/j.dib.2020.105474; J. Yang et al., *MedMNIST v2*, Sci. Data **10**, 41 (2023), doi:10.1038/s41597-022-01721-8

# Blood Cells without Labels




- the 3421 BloodMNIST test images; left to right: PCA of the pixels, t-SNE of the pixels, t-SNE of the 16 latent numbers of a convolutional autoencoder trained on the images alone (cell types only colour the points)
- k-means ($k = 8$) against the cell types, ARI: 0.33 (50 PCA components), 0.34 (latent space)
- platelets and erythroblasts form their own groups; the white-blood-cell types overlap, in the pixels and in the latent space

# t-SNE Settings for Blood Cells




- perplexity 5, 30, 100 on three inputs; score: vote of the 10 nearest neighbours (chance 0.20) and k-means ARI; dotted: the input before t-SNE
- no setting separates the eight types: vote 0.58–0.67 on the maps, 0.65–0.71 on the inputs; ARI at most 0.32
- t-SNE keeps neighbours, and the neighbours are already mixed in the input; possibly pixels mostly carry colour, size and brightness, while the white-cell types differ in finer detail

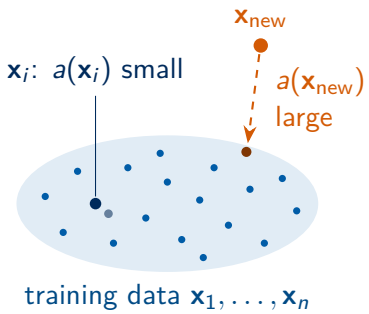
# 7

# Anomaly Detection



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# Anomaly Detection with Reconstruction Error




light dots: reconstructions $\hat{\mathbf{x}}$

Train the autoencoder on the data $\mathbf{x}_1, \ldots, \mathbf{x}_n$:

$$\mathcal{L} = \frac{1}{nD} \sum_{i=1}^{n} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2, \qquad \hat{\mathbf{x}}_i = G_{\theta_D}(E_{\theta_E}(\mathbf{x}_i))$$

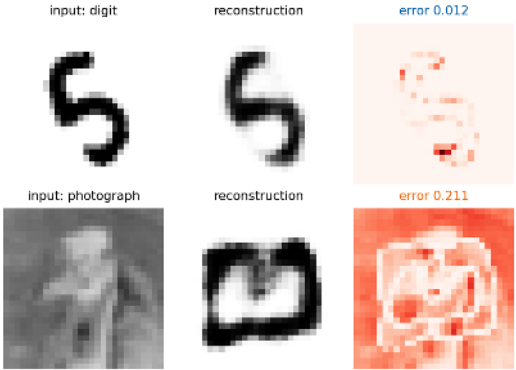
so for typical data $\hat{\mathbf{x}}_i \approx \mathbf{x}_i$. The decoder produces only what it learned: a new $\mathbf{x}$ unlike the training data is mapped back towards them, far from itself.

**Anomaly score** of any $\mathbf{x}$, and the decision:

$$a(\mathbf{x}) = \frac{1}{D} \|\mathbf{x} - G_{\theta_D}(E_{\theta_E}(\mathbf{x}))\|^2,$$

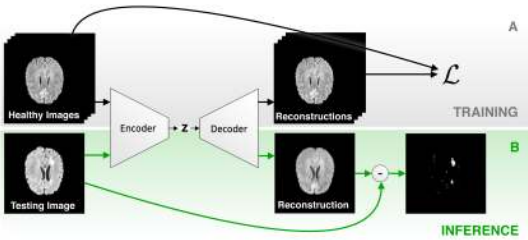
*anomaly* if $a(\mathbf{x}) > \tau$; $\tau$: e.g. exceeded by 5% of normal samples (5% false alarms).

# A Digit and a Photograph




- autoencoder trained on all MNIST digits
- a typical digit: rebuilt well, small $a(\mathbf{x})$
- a photograph: rebuilt as a blurred, digit-like image, large $a(\mathbf{x})$: an **anomaly**
- shown: the cases with the median score of each kind

# Anomaly Detection in Practice

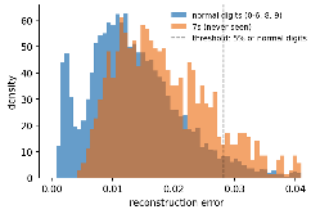



- **medical imaging:** an autoencoder trained on brain MRI of healthy people (A) cannot rebuild a *lesion* (a region of abnormal tissue, e.g. a tumour); scan minus reconstruction shows where it is (B)
- **fraud**: a card transaction that does not fit the usual patterns
- **science**: a faulty measurement

C. Baur et al., *Autoencoders for unsupervised anomaly segmentation in brain MR images: A comparative study*, Med. Image Anal. **69**, 101952 (2021), doi:10.1016/j.media.2020.101952; V. Chandola, A. Banerjee and V. Kumar, *Anomaly detection: A survey*, ACM Comput. Surv. **41**, 15 (2009), doi:10.1145/1541880.1541882

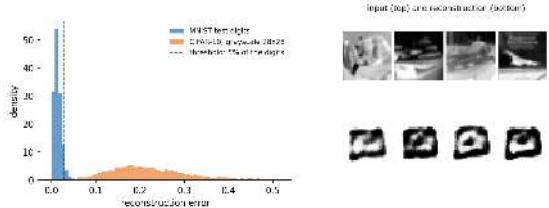
# Two Anomalies

## subtle: the digit 7




- trained without the 7s; blue: trained digits, orange: 7s
- 5% false alarms flag only 13% of the 7s (AUC 0.68), possibly because a 7 shares strokes with 1 and 9

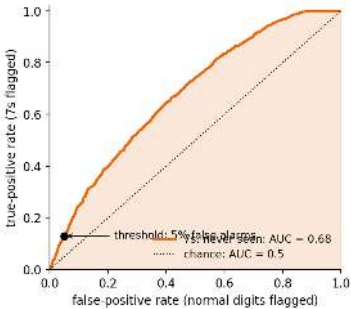
## clear: photographs




- trained on all digits; input: CIFAR-10 photographs, $28 \times 28$ gray
- the same rule flags 99.8% of them (AUC 0.999)

# ROC Curve and AUC

**Purpose:** compare anomaly scores without choosing a threshold first.




$n_A$ anomalies and $n_N$ normal samples; for a threshold $\tau$, $k_A(\tau)$ anomalies and $k_N(\tau)$ normal samples have $a > \tau$:

$$\mathrm{TPR}(\tau) = \frac{k_A(\tau)}{n_A}, \qquad \mathrm{FPR}(\tau) = \frac{k_N(\tau)}{n_N}$$

(anomalies found, false alarms). Lowering $\tau$ traces the **ROC curve** from $(0, 0)$ to $(1, 1)$; chance: the diagonal.

$$\mathrm{AUC} = P\big(a(\mathbf{x}_{\text{anomaly}}) > a(\mathbf{x}_{\text{normal}})\big)$$

area under the curve (random anomaly vs random normal sample): 0.5 for chance, 1 for perfect separation

# Credit-Card Fraud Data

| | V1 | V2 | ⋯ | V28 | ln(1+A) | Class |
|---|---|---|---|---|---|---|
| | -1.36 | -0.07 | ⋯ | -0.02 | 5.01 | 0 |
| | 1.19 | 0.27 | ⋯ | 0.01 | 1.31 | 0 |
| $\mathbf{x}_i$ | -1.36 | -1.34 | ⋯ | -0.06 | 5.94 | 0 |
| | -0.97 | -0.19 | ⋯ | 0.06 | 4.82 | 0 |
| | -1.16 | 0.88 | ⋯ | 0.22 | 4.26 | 0 |
| | | | ⋮ | | | |

$D = 29$ features: $\mathbf{x}_i = (x_{i1}, \ldots, x_{i,29})$

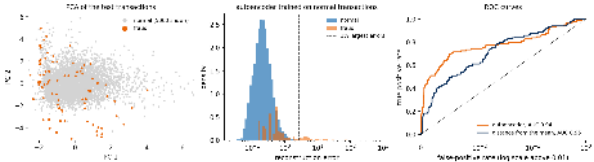
only to evaluate

One row per transaction (first five shown): $\mathbf{x}_i$ is a vector of 29 numbers.

- 284 807 transactions of European cardholders, two days in September 2013; 492 frauds (0.17%)
- `V1`–`V28`: principal components of original features not released (confidentiality), so no readable meaning; $A$: amount
- `Class` (1 = fraud): only to evaluate

A. Dal Pozzolo, O. Caelen, R. A. Johnson, G. Bontempi, *Calibrating Probability with Undersampling for Unbalanced Classification*, IEEE SSCI 2015, 159, doi:10.1109/SSCI.2015.33; data: OpenML 1597

# Fraud as an Anomaly



- **PCA:** two components keep 10% of the variance; many frauds lie at the edge of the cloud, many inside it
- **autoencoder** ($29 \to 16 \to 8 \to 16 \to 29$, trained on normal transactions): flagging the 1% largest errors catches 73 of 98 frauds (74%); 12.8% of the flagged transactions are fraud
- **ROC:** both scores have AUC $\approx 0.95$; at 0.1% false alarms the autoencoder finds 44% of the frauds, the distance 18%

A bank can review only a small share of transactions, so what counts is the fraction of frauds found when very few normal ones are flagged.

# Tumour Gene Expression Data

| | gene_1 | gene_2 | gene_3 | ⋯ | gene_20530 | tumour type |
|---|---|---|---|---|---|---|
| | 2.02 | 3.27 | 5.48 | ⋯ | 0.00 | PRAD |
| $\mathbf{x}_i$ | 0.59 | 1.59 | 7.59 | ⋯ | 0.00 | LUAD |
| | 3.51 | 4.33 | 6.88 | ⋯ | 0.00 | PRAD |
| | 3.66 | 4.51 | 6.66 | ⋯ | 0.00 | PRAD |
| | 2.66 | 2.82 | 6.54 | ⋯ | 0.00 | BRCA |
| | | | ⋮ | | | |

$D = 20\,531$ genes

only to evaluate

One row per tumour sample: $\mathbf{x}_i$ = expression level of each gene (RNA sequencing), anonymised genes, values 0–20.8 (apparently a log scale; normalisation not documented). Tumour type: a label, used only to evaluate.

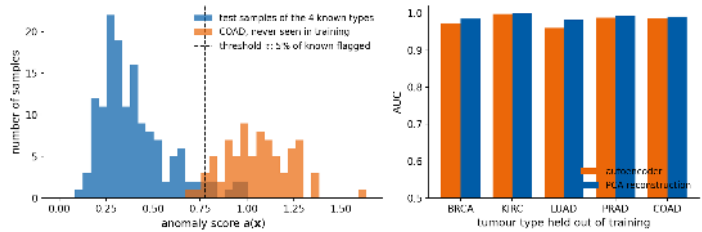
| | |
|---|---|
| BRCA | breast invasive carcinoma |
| KIRC | kidney renal clear cell carcinoma |
| LUAD | lung adenocarcinoma |
| PRAD | prostate adenocarcinoma |
| COAD | colon adenocarcinoma |

801 samples from The Cancer Genome Atlas (TCGA): BRCA 300, KIRC 146, LUAD 141, PRAD 136, COAD 78.

S. Fiorini, *gene expression cancer RNA-Seq*, UCI Machine Learning Repository (2016), doi:10.24432/C5R88H;

J. N. Weinstein et al., *The Cancer Genome Atlas Pan-Cancer analysis project*, Nat. Genet. **45**, 1113 (2013), doi:10.1038/ng.2764.

# An Unseen Tumour Type as an Anomaly




- train on four types (80%); 2000 most variable genes, chosen and standardised on the training part only; autoencoder $2000 \to 256 \to 32 \to 256 \to 2000$
- COAD held out: $\tau$ flags 5% of the known test samples and 96% of the COAD samples; each type held out in turn: AUC 0.96–1.00, PCA reconstruction (32 components) 0.98–1.00
- the types differ in many genes: an unseen type lies far from the training data, an easy anomaly that PCA already finds

# Methods without Labels

| method | gives | new data | keeps / limits |
|---|---|---|---|
| PCA | linear projection, largest variance | projected with $V_m$ | global structure; misses curved structure |
| t-SNE | a 2D map for looking | no: the map is refitted | neighbours; gaps and cluster sizes are not distances |
| k-means | $k$ groups with centres | assigned to the nearest centre | round groups of similar size; $k$ must be chosen |
| DBSCAN | groups of any shape, plus noise | not directly | dense regions; needs the radius $\varepsilon$ and $m$ |
| autoencoder | encoder, decoder, latent space | encoded and rebuilt | curved structure; reconstruction error flags anomalies |

Common pipeline: PCA to a few tens of dimensions, then t-SNE or clustering; an autoencoder for curved structure or anomalies.

# Summary

- **unsupervised learning** finds structure in data without labels: a few coordinates, groups, unusual samples
- **PCA**: a linear projection on the directions of largest variance; **t-SNE**: a non-linear map that keeps neighbours
- **k-means**: groups around $k$ centres, $k$ chosen by us (elbow, silhouette); it assumes round clusters; **DBSCAN** follows dense regions of any shape
- **autoencoder**: compress to a latent space and rebuild; the reconstruction error flags unusual samples, judged by the ROC curve and AUC
- **tables**: the same tools on card transactions and on tumour gene expression; an unseen tumour type is an anomaly

# Next Lecture: Learning a Probability Distribution from Data

The data $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are samples from an unknown distribution $p(\mathbf{x})$. Can we learn $p$ itself, and draw *new* samples from it?

- probability, maximum likelihood and latent-variable models
- variational autoencoders: a latent space we can sample, new digits from random latent vectors
- conditional generation: a chosen digit or cell type
- disentangled latent features ($\beta$-VAE) as a tool for interpretation

**The latent space of an autoencoder becomes a probability distribution we can draw new data from.**

# Thank You!

Questions, project ideas, feedback: marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar

